

Slamming: Recent Progress in the Evaluation of Impact Pressures

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Abstract

Slamming, the violent impact between a liquid and solid, has been known to be important for a long time in the ship hydrodynamics community. More recently, applications ranging from the transport of liquefied natural gas (LNG) in LNG carriers to the harvesting of wave energy with oscillating wave surge converters have led to renewed interest in the topic. The main reason for this renewed interest is that the extreme impact pressures generated during slamming can affect the integrity of the structures involved. Slamming fluid mechanics is challenging to describe, as much from an experimental viewpoint as from a numerical viewpoint, because of the large span of spatial and temporal scales involved. Even the physical mechanisms of slamming are challenging: What physical phenomena must be included in slamming models? An important issue deals with the practical modeling of slamming: Are there any simple models available? Are numerical models viable? What are the consequences for the design of structures? This article describes the loading processes involved in slamming, offers state-of-the-art results, and highlights unresolved issues worthy of further research.



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LNG: liquefied natural gas
OWSC: oscillating wave surge converter
ELP: elementary loading process

1. INTRODUCTION

Slamming, the violent impact between a liquid and solid, has been known for a long time to be an important aspect of ship hydrodynamics and seaplane mechanics. Von Kármán (1929) published a paper on the estimation of forces on the floats of landing seaplanes. The foundational paper on the problem of slamming (Wagner 1932) was written 85 years ago but is surprisingly modern. The various applications of slamming are in aeronautics (floats of landing seaplanes and helicopters), shipbuilding [the acceptable amount of bow flare angle for use at sea, stern slamming, high-speed vessel slamming, wet-deck slamming (see Korobkin 1998, Kapsenberg 2011)], the liquid gas industry [containment systems for liquefied natural gas (LNG) (see Faltinsen & Timokha 2009, Int. Ship. Offshore Struct. Congr. 2015)], and, more recently, marine renewable energy [flaps of oscillating wave surge converters (OWSCs) (see Henry et al. 2014b) and offshore wind turbines].

But the range of applications is much larger. Indeed, the liquid impacts that occur inside of the tanks of LNG carriers can be studied as waves breaking against a vertical wall, which is likewise of great importance to ocean engineering and was reviewed by Peregrine (2003). For decades, researchers and engineers have studied the violent impacts of breaking waves on breakwaters and sea walls built to protect areas of water and land from the action of water waves (Blackmore & Hewson 1984, Kirkgöz 1990, Oumeraci et al. 1993, Hattori et al. 1994, Bullock et al. 2001, Cuomo et al. 2010). The idealized problem used in the aeronautics and shipbuilding industries is the water entry problem. The main difference between the water entry problem and the problem of a wave breaking against a wall is that in the former the kinetic energy results from the fast-moving solid, whereas in the latter the kinetic energy results from the fast-moving liquid. Quite surprisingly, the existing literature is strongly divided between the two problems. For instance, Peregrine's (2003) review, titled "Water-Wave Impact on Walls," does not mention slamming or water entry. In his article "Slamming of Ships: Where Are We Now?," Kapsenberg (2011, figure 4) referred only once to the flat impact of a breaking wave against a vertical wall. However, the problem of a breaking wave against a wall is as old as the water entry problem: Indeed, in the same way that Wagner's (1932) paper is considered the foundational paper for the water entry problem, Bagnold's (1939) paper is considered the foundational paper for the problem of a wave impacting a wall, even though Bagnold originally studied only the compression of a gas pocket by a liquid. The initial stage of the impact of a jet on a solid boundary has been much less studied because researchers must take the compressibility of the liquid into account to properly describe the shock waves generated inside the liquid by the impact. Studies of the root mechanisms have origins even further back and are closely related to the Rankine–Hugoniot conditions (Rankine 1870; Hugoniot 1887, 1889). The corresponding canonical models illuminate the physical phenomena involved in slamming.

The rest of Section 1 describes the concept of elementary loading processes (ELPs), provides the context of the various applications, and reviews the state of the art. Section 2 reviews the mathematical models, and Section 3 reviews their solutions. Section 4 summarizes the experimental techniques, whereas Section 5 discusses the consequences of slamming on the design of structures. Finally, a list of open questions is provided. The present review focuses on impacts with bouncing effects as opposed to impacts for which the solid motion continues after impact; other aspects of free-surface impact, such as the water entry of projectiles and cavity formation (Korobkin 1997, Truscott et al. 2014), are not discussed. Hydroelasticity effects are also omitted from the present review (Faltinsen 2000, Malenica et al. 2009, el Moctar et al. 2011, Ten et al. 2011).

1.1. Elementary Loading Processes

A slamming event is characterized by a sudden high force of relatively short duration imposed on a body. This event occurs when a body hits a fluid or vice versa with a small relative angle between

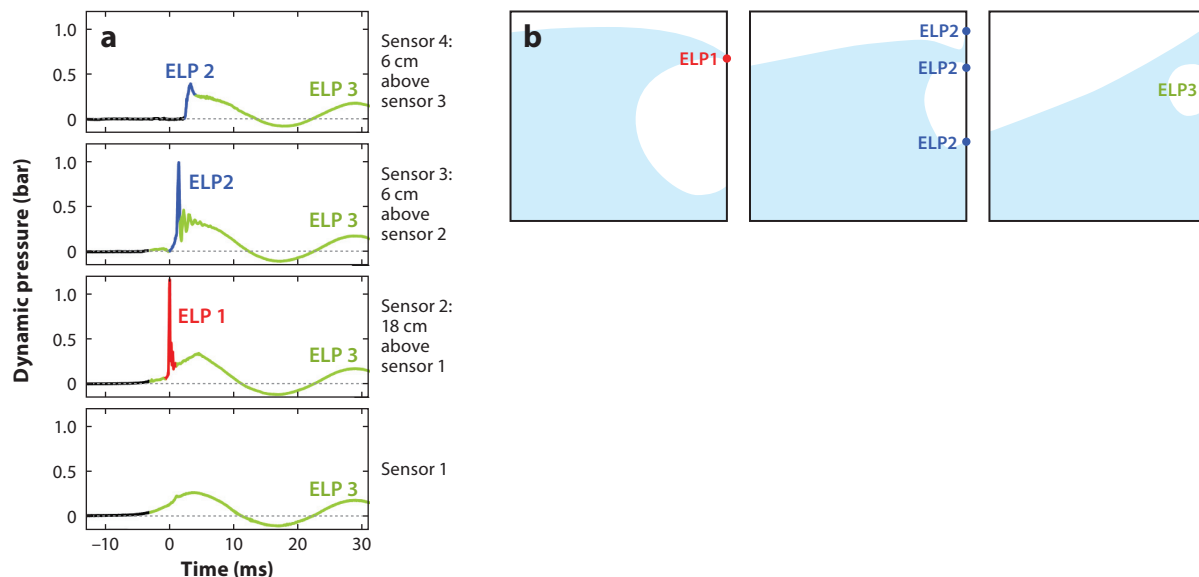


Figure 1

(a) Elementary loading processes (ELPs) illustrated through the pressure history following a wave impact on a wall obtained in the framework of the Sloshe experiments (Lafeber et al. 2012a). The dynamic pressure is the pressure minus ~ 1 bar, the ambient pressure. The contribution of each ELP is indicated: direct impact (ELP1, red), building jet (ELP2, blue), and compression of entrapped or escaping gas (ELP3, green). Adapted with permission from Lafeber et al. (2012a). (b) Qualitative descriptions of the three ELPs, with the same color coding for the circles as in panel a.

the body surface and fluid surface. The contact region between the fluid and solid expands much more rapidly than the impact speed.

Lafeber et al. (2012b) introduced the concept of ELPs to describe the loading processes. Three ELPs have been identified as the building blocks of any load possibly induced by a wave impact. They are the direct impact (ELP1), the building jet along the structure (ELP2), and the compression of entrapped or escaping gas (ELP3), characterized by a pulsating load with a loaded area depending on the amount of entrapped gas. **Figure 1** shows an example of measured pressure history along a wall, with the contribution of each ELP indicated.

The most common ELP is the ELP2. It is present as soon as there is run up (or down) of the liquid along the wall. This run-up process induces hydrodynamic loads due to the change of fluid momentum. This is the ELP that also occurs in the water entry problem, studied experimentally in the context of drop tests, described by Wagner (1932). This is also the only ELP that can be satisfactorily reproduced in numerical simulations based on incompressible fluids, and it is by far the most thoroughly described ELP in the literature. The flip-through phenomenon, introduced by Cooker & Peregrine (1990) and illustrated in **Figure 2**, can also be linked to ELP2 if viewed in a frame of reference attached to the wave front. The water level at the wall accelerates rapidly, water close to the wall accelerates even faster, and a vertical jet forms. Very high pressures are reached.

The oscillating air enclosure ELP3 was first described by Bagnold (1939). ELP3 is not as common as ELP2 because additional conditions must be satisfied for it to occur. In the case of the water entry problem, it might occur only if the relative angle between the body and fluid is smaller than 5° . In the case of a breaking wave hitting a vertical wall, it might occur if the crest of the wave hits the wall. Then the forward movement of the wave front combined with the run up of the wave trough leads to the compression of a gas pocket. In turn, the gas pocket acts like a

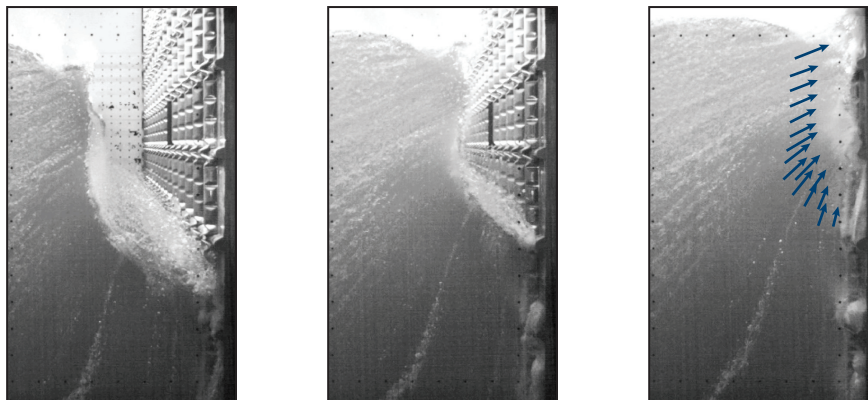


Figure 2

Observation of the flip-through impact with a high-speed camera. (*Right*) Wave shape just before the impact. The blue arrows represent the velocities of the bubbles inside the water near the free surface. Figure reproduced with permission from Brosset et al. (2011).

spring. The maximum pressure inside the pocket and the frequency of the oscillations depend on the size of the pocket.

The direct impact ELP1 has been the least studied, even though it was mentioned by Rankine (1870) and Hugoniot (1887, 1889). It occurs when a solid stops a liquid: The velocities of the liquid particles that touch the solid decelerate rapidly. This near discontinuity in velocity at a contact point generates a hemispherical pressure wave inside the liquid and a hemispherical strain wave inside the solid, both centered on the contact point. The pressure waves propagate at the speed of sound in the liquid.

Clearly, ELP1 requires liquid compressibility, and ELP3 involves gas compressibility. Because bubbles greatly reduce the pure liquid sound speed, the propagation of the impact in ELP1 can slow down considerably if the liquid is aerated, and thus the bubbles lead to a longer-duration impact. In general, liquid compressibility can be neglected if $Ma_\ell^2 < 0.1$, where the Mach number Ma_ℓ is the liquid velocity/speed of sound. Thus, even a liquid with $Ma_\ell = 0.3$ can be considered incompressible.

The combination of ELPs, which can occur, for example, when a gas pocket wave impact occurs on a flat wall, is schematically represented in **Figure 3**, with the labeled submechanisms briefly described in Section 1.3.

1.2. Applications

Kapsenberg's (2011) review described the importance of slamming in shipbuilding, and Faltinsen et al. (2004) and Korobkin & Pukhnachov (1988) described applications to ocean structures. Here it is important to distinguish between fatigue damage and extreme loading. Slamming can introduce a whipping response by the ship's hull, leading to high accelerations and high-frequency bending moments. Severe slamming can cause damage from a single impact. Although extreme loading on the hulls of high-speed vessels is a concern, it is usually not the local peak pressure but net force impulse on the whole impact area that is most important for ship slamming (el Moctar et al. 2011, Kapsenberg 2011).

However, extreme impact pressures are important in cases: for LNG carriers and, although they are at their infancy, for some types of wave energy converters (WECs). Gavory & de Sèze

Mach number: ratio between the flow velocity and local speed of sound;
 $Ma = V/c$

WEC: wave energy converter

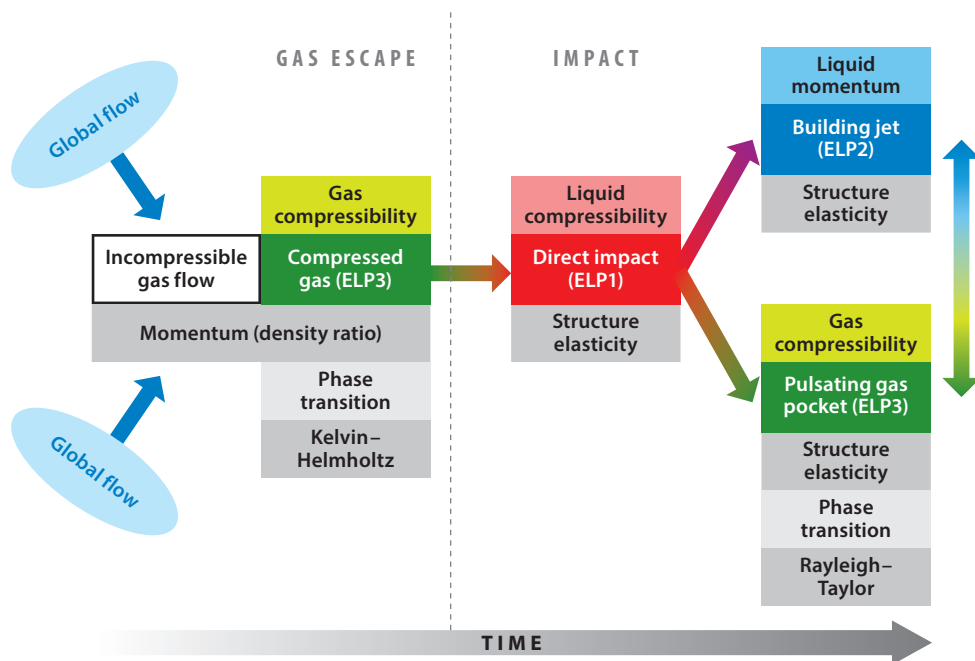


Figure 3

Schematic representation of the most typical combination of elementary loading processes in relation to their physical phenomena. Figure adapted with permission from Lafeber et al. (2012b).

(2009) reported examples of structural damage to tanks of LNG carriers. The slamming of an OWSC can occur in extreme sea states, as the flap pitches seaward and reaches a vertical position in the wave trough: The waterline suddenly drops in front of the flap, and the flap hits the water surface with high angular velocity, leading to a short duration but high-magnitude load on the device. Although there are similarities between the slamming of an OWSC and that of a ship, the former has some unique features. It is also different from wave impact on a stationary wall, in which the pressure distribution on the wall is dominated by the front shape of the wave. Additionally, it is distinct from the water entry problem, for which the initial contact angle and the relative motion are known in advance. There are no available analytical solutions or empirical data that one can directly employ for design.

1.3. State of the Art

In their book on sloshing, Faltinsen & Timokha (2009) included a chapter on slamming (chapter 11), focusing on analytical models and scaling, although they also included experiments and numerical simulations. During the past decade, there have been several new developments in our understanding and capabilities for numerical simulation of slamming events. Researchers have also conducted extensive experimental campaigns, both in academic and industrial environments. Karimi et al. (2015) confirmed that the global flow can be independent of both the scale and density ratio (DR) between the ullage gas and the liquid in the tank.

In contrast, the local flow depends on the DR, so impact pressures will vary. Phase transition (liquid–gas) and surface tension also play a role locally (Ancellin et al. 2012, 2016). Surface tension affects the shape of the interface between the liquid and gas, and the modification of the interface

DR: density ratio between the gas and the liquid

Ullage gas: gas in the space not filled by the liquid in a liquefied natural gas tank

geometry poses great challenges: Small gas pockets can form, and the aeration of the liquid enhances its compressibility. The problem of liquid impact on a wall, such as described by Faltinsen & Timokha (2009), has become the primary benchmark for the study of slamming. Several groups have been using the extreme waves computed with potential flow theory by Scolan (2010) as initial conditions for a variety of compressible multiphase flow solvers to study the impact on a vertical wall (Bredmose et al. 2009; Guilcher et al. 2013, 2014a; Behruzi et al. 2014; Costes et al. 2014; Rafiee et al. 2015). Scalings to describe the phenomena have also been developed and tested.

Though the global scaling is well understood, scaling remains poorly understood at the local level. Experiments at different scales have revealed that the shape of the interface, for example, does not scale well. The gas–liquid DR modifies wave shapes prior to wave breaking (Karimi et al. 2016). The instabilities at the interface can also be important. Frihat et al. (2016) showed that the main sources of variability for the flow are the development of free-surface instabilities just before an impact, the fall of droplets on a large width of the interface just after any impact splashing, and the production of bubbles. Fortunately, other phenomena regulate the overall flow, such as the forced motions imposed by the sloshing rig and the dissipation.

2. MATHEMATICAL MODELS

The analysis of both external loads on the hull and internal loads on the tank walls for LNG carriers is usually performed in the frequency domain, using three-dimensional (3D) hydrodynamic models. For a set of wave frequencies, transfer functions (i.e., response amplitude operators) are obtained by running hydrodynamic simulations for a wave of unit amplitude. This requires a number of simplifying assumptions, including linearity. Response amplitude operators provide wave-induced global loads; motions, velocities, and accelerations; dynamic pressures; and wave kinematics. They also include slamming occurrences although, in this case, the underlying assumptions are questionable. More realistic analyses employ small-scale experimental models supported by extensive developments of CFD techniques and focus on one isolated tank or WEC subjected to forced motion. For OWSCs, the mechanism is more subtle because large flap-type absorbers are driven by diffraction (Renzi et al. 2014). The models introduced in the following sections concentrate on these local flows (however, see Karimi et al. (2015), who introduce the concept of impact coincidence), described in Section 1.3.

2.1. The Most General Fluid Model

The most general model is that of the Navier–Stokes (NS) equations for the barotropic 3D flow of two immiscible compressible fluids with surface tension. Time is represented by the variable t , and the space variables are denoted by $\mathbf{x}$. Velocities are denoted by $\mathbf{v}(t, \mathbf{x})$ and pressure by $p(\mathbf{x}, t)$.

2.1.1. Geometry of the interface between the two fluids. We assume that the two compressible fluids are separated by a smooth interface $\Sigma(t)$. We take

$$\varphi(t, \mathbf{x}) = 0 \quad 1.$$

to be the interface, and we assume that $\varphi < 0$ corresponds to the gas and $\varphi > 0$ corresponds to the liquid (**Figure 4**).

We denote by $\mathbf{n}$ the unit normal vector to the free surface at time t :

$$\mathbf{n} = \frac{\nabla_{\mathbf{x}} \varphi}{|\nabla_{\mathbf{x}} \varphi|}, \quad 2.$$

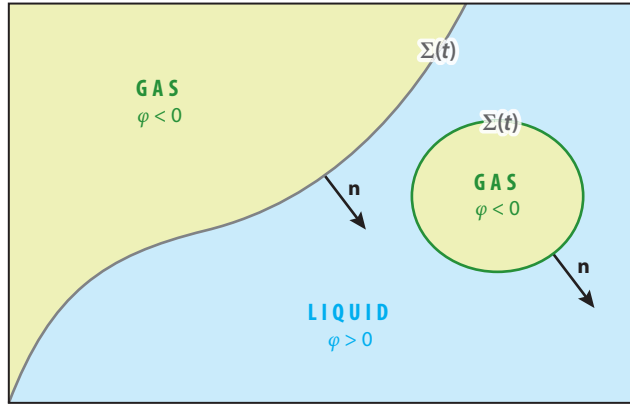


Figure 4

The interface $\Sigma(t)$ at time t between the gas and liquid is represented by the line and the circle. The interface can be fragmented. Along the interface, $\varphi(t, \mathbf{x}) = 0$, with $\varphi < 0$ corresponding to the gas and $\varphi > 0$ to the liquid. $\mathbf{n}$ is the unit normal vector to the free surface at time t .

with the free surface given by

$$\Sigma(t) \equiv \{\mathbf{x} \in \mathbb{R}^3 \text{ such that } \varphi(t, \mathbf{x}) = 0\}. \quad 3.$$

2.1.2. Local conservation laws. For the general model considered in this review, mass conservation and momentum balance are written, respectively, as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad 4.$$

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} - \boldsymbol{\sigma}) = \rho \mathbf{g} + 2 \omega \kappa \delta_{\Sigma} \mathbf{n}, \quad 5.$$

where $\mathbf{g}$ is the acceleration due to gravity; ρ is the fluid density; $\boldsymbol{\sigma}$ is the stress tensor; and $2 \omega \kappa \delta_{\Sigma} \mathbf{n}$ is the capillary force between both fluids, with ω the surface tension coefficient, κ the mean curvature of the interface Σ , and δ_{Σ} the Dirac distribution localized on the interface.

2.1.2.1. Equation of state and constitutive relation. In general, to include phase transition, one also needs an energy balance equation. However, for the sake of simplicity, this is neglected here and a stiffened extension of the ideal gas equation of state (EoS) is used instead:

$$\rho = \rho^{\text{ref}} \left(1 + \gamma \chi \frac{p - p^{\text{ref}}}{p^{\text{ref}}} \right)^{1/\gamma}, \quad 6.$$

where ρ^{ref} and p^{ref} are reference values, and γ and χ are the adiabatic constant and the compressibility factor of the fluid, respectively.

The ideal gas EoS is recovered for $\chi = 1/\gamma$, and $\chi = 0$ yields an incompressible fluid ($\rho = \rho^{\text{ref}}$). Hence, Equation 6 can be considered an interpolation between these two cases and is referred to as the isentropic stiffened gas EoS. For a gas, χ is of order one (e.g., $\chi = 5/7$ for air in ordinary conditions), whereas $\chi \ll 1$ for a liquid (e.g., $1/\chi \simeq 22,500$ for water in normal conditions). For $\chi \ll 1$, Equation 6 is well approximated by

$$\rho = \rho^{\text{ref}} \left(1 + \chi \frac{p - p^{\text{ref}}}{p^{\text{ref}}} \right). \quad 7.$$

EoS: equation of state

The fluids are Newtonian so that the stress tensor $\boldsymbol{\sigma}$ can be written as

$$\boldsymbol{\sigma} = -p \mathbf{I} + 2\eta \mathbf{D} - \frac{2\eta}{3} (\nabla \cdot \mathbf{v}) \mathbf{I}, \quad 8.$$

where

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{v} + (\nabla \mathbf{v})'], \quad 9.$$

and η is the dynamic viscosity.

2.1.2.2. Initial and boundary conditions. Initially, both the density and velocity of the fluids are known, as well as the geometry of the interface between both fluids.

There are two kinds of boundary conditions. First, on the boundary walls (assumed to be fully rigid; elasticity is not considered in this review), we write the usual no-slip condition:

$$\mathbf{v}(t, \mathbf{x}) = \mathbf{0}. \quad 10.$$

Second, on the interface $\Sigma(t)$ between the two fluids (with g the gas and ℓ the liquid), we write the kinematic condition,

$$\varphi_t + \mathbf{v}_k \cdot \nabla_x \varphi = 0, \quad k = g \text{ or } k = \ell, \quad 11.$$

the continuity of velocity across $\Sigma(t)$,

$$[\mathbf{v}] = \mathbf{0}, \quad 12.$$

and the dynamic condition,

$$[\boldsymbol{\sigma} \cdot \mathbf{n}] = 2 \omega \kappa \mathbf{n}. \quad 13.$$

2.2. Averaged Models

When the geometry of the interfaces is complicated (e.g., if there is the formation of bubbles or droplets), it is almost impossible to track them accurately by numerical methods, which motivates so-called averaged models. Readers are referred to Ishii & Hibiki's (2011, chapter 3) extensive discussion of averaging. All averaging methods lead to a set of equations that have a common mathematical structure:

$$\frac{\partial(\alpha_g \rho_g)}{\partial t} + \nabla \cdot (\alpha_g \rho_g \mathbf{v}_g) = 0, \quad 14.$$

$$\frac{\partial(\alpha_\ell \rho_\ell)}{\partial t} + \nabla \cdot (\alpha_\ell \rho_\ell \mathbf{v}_\ell) = 0, \quad 15.$$

$$\frac{\partial(\alpha_g \rho_g \mathbf{v}_g)}{\partial t} + \nabla \cdot (\alpha_g \rho_g \mathbf{v}_g \otimes \mathbf{v}_g - \alpha_g \boldsymbol{\sigma}_g) = \alpha_g \rho_g \mathbf{g} + \mathbf{M}_g, \quad 16.$$

$$\frac{\partial(\alpha_\ell \rho_\ell \mathbf{v}_\ell)}{\partial t} + \nabla \cdot (\alpha_\ell \rho_\ell \mathbf{v}_\ell \otimes \mathbf{v}_\ell - \alpha_\ell \boldsymbol{\sigma}_\ell) = \alpha_\ell \rho_\ell \mathbf{g} + \mathbf{M}_\ell, \quad 17.$$

where α_g and α_ℓ are the volume fraction of the gas and liquid, respectively, and $\mathbf{M}_g$ and $\mathbf{M}_\ell$ denote the exchange of momentum between the fluids. They satisfy

$$\alpha_g + \alpha_\ell = 1 \text{ and } \mathbf{M}_g + \mathbf{M}_\ell = \mathbf{0}. \quad 18.$$

Notice that both the EoS and the constitutive relation are written for each fluid. The central challenge in averaging is closure of exchange terms $\mathbf{M}_g$ and $\mathbf{M}_\ell$. This is beyond the scope of the present review, and we refer readers to Ishii & Hibiki (2011). We simply mention that the closure relation depends on the so-called regime (separated, transitional, or dispersed) under consideration. Moreover, in each regime, the closure relation will depend of the type of flow. For example, when dealing with dispersed flows, one has to address either bubbly flow or droplet flow.

Once the model has been agreed upon, the next step is to choose a numerical method to produce a discrete version of the continuous equations. This is discussed in Section 3.2.2.

2.3. Compressible Aerated Flow Models

Alternatives to the above general system of Equations 14–18 and 6 are the so-called compressible aerated flow models. These models accommodate both the entrapment and entrainment of gas by including a compressible gas fraction within the liquid. The aerated fluid is considered to be a homogeneous mixture of incompressible liquid and compressible gas (Bredmose et al. 2009, 2015; Dias et al. 2010). It is also assumed that there is only one velocity field ($\mathbf{v}_g = \mathbf{v}_\ell \equiv \mathbf{v}$). Adding Equations 16 and 17 yields

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} - \boldsymbol{\sigma}) = \rho \mathbf{g}, \quad 19.$$

$$\rho \equiv \alpha_g \rho_g + \alpha_\ell \rho_\ell, \quad \boldsymbol{\sigma} \equiv \alpha_g \boldsymbol{\sigma}_g + \alpha_\ell \boldsymbol{\sigma}_\ell. \quad 20.$$

We observe that the possibility of pressure shock waves makes the above pressure description incomplete. In addition to ensuring the conservation of mass and momentum, one must ensure the conservation of total energy to accommodate shock waves. This allows the entropy to increase owing to a shock wave, whereas Equation 6 enforces the conservation of entropy (isentropic flow).

2.4. Some Incompressible Fluid Models

As indicated in Section 1, incompressible fluid models can successfully describe some aspects of slamming, such as ELP2 and the flip-through phenomenon. Therefore, we provide a brief review of such models here. The general model, or even the averaged models described in the previous sections, can be simplified in several different ways. One is to consider one- or two-fluid formulations of the unsteady incompressible NS equations with an interface and to add a turbulence model. This is the simplified model used in multipurpose codes such as ANSYS Fluent or OpenFOAM. A popular OpenFOAM solver is interDyMFoam, which is a solver for two incompressible, isothermal, immiscible fluids. (The set of OpenFOAM and ANSYS Fluent solvers is constantly growing, and there are now compressible multiphase flow solvers; one of them was used in the numerical comparative study described in Section 3.2.6.) The momentum equation (Equation 5) is replaced by a similar equation with an effective viscosity that includes the molecular viscosity and turbulent viscosity. Particle methods such as the incompressible smoothed particle hydrodynamics (ISPH) method have also been used. The main advantage of the ISPH method in simulating violent free-surface flows is its ability to deal with large deformations and fragmentations (see Gotoh et al. 2014). The most challenging issue with the ISPH method is the presence of unphysical pressure fluctuations. Not surprisingly, several studies have recently been devoted to the development of enhanced schemes to predict impact pressures.

3. SOLUTIONS TO THE MODELS

This section describes some results obtained with the models introduced above. Asymptotic solutions, which are presented first, are used to isolate the physical phenomena that can occur during slamming. They also provide order of magnitude estimates of the loads. Then more advanced models (including experiments or numerical methods) are introduced to provide additional details and identify the limits of the asymptotic theories (see the sidebar titled Compressible Versus Incompressible Fluid Models for a short discussion on which model to use).

COMPRESSIBLE VERSUS INCOMPRESSIBLE FLUID MODELS

Compressible fluid models require more computing time than incompressible fluid models. So it is important to know in which situations compressible fluid models are absolutely necessary. Surprisingly, potential flow models can be quite useful for slamming problems. In fact, the majority of numerical work devoted to the water entry problem and to the flip-through phenomenon is based on potential flow theory. But the main limitation of the incompressible flow models for the study of slamming is that they cannot deal with ELP1 and ELP3, because the compressibility of the liquid and/or the gas becomes important. When ELP2 dominates the violent impact, incompressible flow models can be used. In a situation where either ELP1 or ELP3 is present, there are several important differences in the results if one uses an incompressible flow model as opposed to a compressible flow solver. The first difference is the time of the impact. The impact occurs sooner with the incompressible model because the liquid goes faster. The gas pockets drain very quickly because there is no limit on the gas speed. The second difference is that the magnitude of the maximum pressure is larger with the compressible model because of the compressibility of the liquid which leads to ELP1 (Figure 5). An interesting alternative is to couple an incompressible fluid model with a compressible fluid model only when compressibility is needed (see Section 3.2.6) (Costes et al. 2014, Mrabet 2017).

3.1. Asymptotic Solutions

In this section, we describe three asymptotic solutions corresponding to the three main slamming configurations: Wagner's (1932) solution (ELP2), Bagnold's (1939) solution (ELP3), and Rankine–Hugoniot solution (ELP1) (Rankine 1870; Hugoniot 1887, 1889). Each analysis leads to the scaling laws for the particular phenomenon. Readers are referred to Braeunig et al.'s (2009a) synopsis of scaling in this context. A more complete study is provided by L. Brosset & J.M. Ghidaglia (unpublished manuscript). By scaling, we mean the following. Let us assume that the

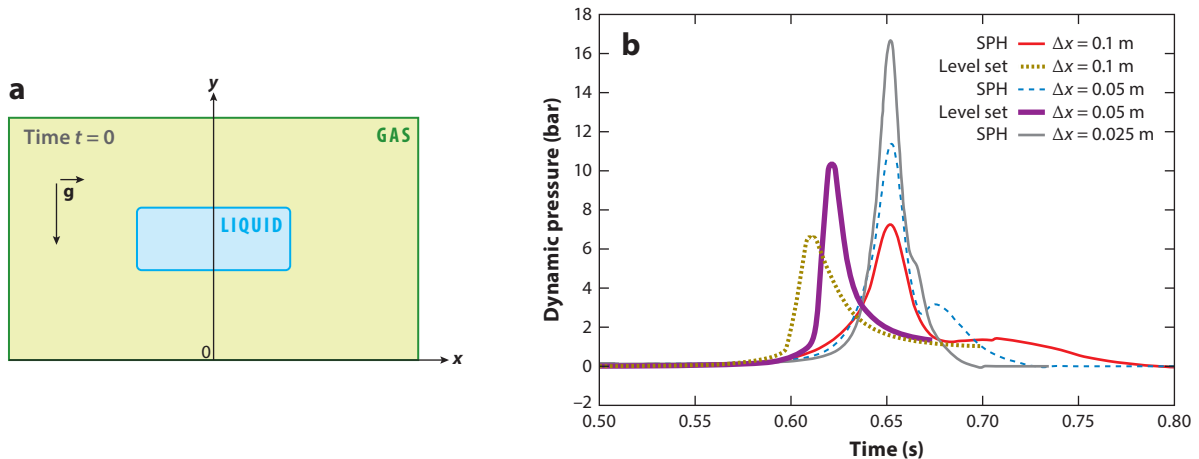


Figure 5

(a) Example of a liquid patch of rectangular shape falling freely within a two-dimensional rectangular tank filled with gas. Adapted with permission from an International Society of Offshore and Polar Engineers numerical comparative study initiated by Braeunig et al. (2009a) and conducted in 2010 with an update in 2013. (b) The maximum impact pressure is smaller and arrives earlier with an incompressible model (using the level set method to track the interface) than with a compressible model (SPH; see Section 3.2.5). Adapted with permission from Rafiee et al. (2015). Abbreviation: SPH, smoothed particle hydrodynamics.

PARTIAL FROUDE SCALING

Discussions in Section 3.1 of the scaling for the Bagnold (1939) and Rankine–Hugoniot solutions are based on the fact that the same fluids are used for the full-scale prototype and for the model, or at least that both fluids have similar speeds of sound. When the compressibility of the gases and liquids is not adequately scaled at both scales, the conditions are referred to as partial Froude scaling (PFS) conditions (Braeunig et al. 2009a). For instance, this is the case when the fluids remain the same at both scales. In that case, there are gas and liquid compressibility biases that spoil Froude similarity. If the compressibility of the gases and liquids was scaled according to Froude scaling, then the pressure would scale as the full-to-model scale ratio λ , like in the incompressible case. We then talk about complete Froude scaling.

ratio of the full-to-model scale is λ . The key question is, How do slamming-induced pressures scale? That is, what is the ratio between $p_{\max}^{\text{full scale}}$ and $p_{\max}^{\text{model}}$? An important dimensionless number is expected, of course, to be the Froude number Fr :

$$\text{Fr} = \frac{V}{\sqrt{gL}}, \quad 21.$$

where V and L are a characteristic velocity and characteristic length, respectively. Preserving the Froude number at both scales and keeping in mind that the acceleration due to gravity is the same at both scales imply that the full-scale velocity must be $\sqrt{\lambda}$ times the model velocity or, equivalently, that time in the model scales as $1/\sqrt{\lambda}$ with respect to the full-scale time. In situations where compressibility plays a role, Froude similarity should in theory also apply to the speeds of sound. However, this is not possible in reality, leading to compressibility biases (see the sidebar titled Partial Froude Scaling).

3.1.1. Wagner's solution. Wagner's (1932) solution describes the fall of a wedge at a velocity V (Figure 6). It uses potential flow theory and is based on a linearization of both the wetted and free surfaces that are projected on the initial free surface at rest. The determination of the contact point then reduces to the so-called Wagner condition, which prescribes the continuity of the displacement between fluid particles lying on the free and wetted surfaces. The main result of

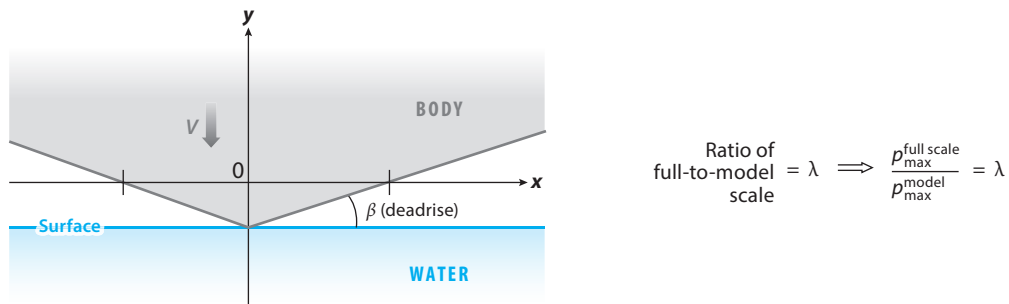


Figure 6

Two-dimensional water entry problem (elementary loading process ELP2). A free-falling wedge hits the free surface of a liquid initially at rest with a vertical velocity V . The deadrise angle β is the angle between the free and wetted surfaces. For a full-to-model scale ratio λ , the full-to-model pressure ratio $p_{\max}^{\text{full scale}} / p_{\max}^{\text{model}}$ is also λ .

Wagner's solution is that the impact pressure is

$$p_{\max} = \frac{1}{2} C_p \rho_\ell V^2, \quad 22.$$

with $C_p = \pi^2/(4 \tan^2 \beta)$, where ρ_ℓ is the liquid density and β is the deadrise angle (i.e., the angle between the free and wetted surfaces) (**Figure 6**). With Froude scaling and the same liquid density, the pressure $p_{\max}$ due to ELP2 scales as λ . We note the limitation that $p_{\max}$ tends to infinity as the deadrise angle goes to zero. More generally, $p_{\max}$ is directly related to the square of the normal velocity of the expanding wetted surface (Scolan & Korobkin 2003).

The generalized Wagner model (Zhao et al. 1997), which only requires the free surface to be linearized, represents a good trade-off between the linearized Wagner model and a fully nonlinear approach, still based on potential flow theory. This linearization makes the model less physical than the full one, but the exact definition of the wetted surface removes some of the limitations of the linearized Wagner model. Xu et al. (2010) and Sun et al. (2015) have proposed generalizations to free falls with more degrees of freedom. For an extensive coverage of Wagner's (1932) solution, readers are referred to Faltinsen & Timokha (2009). The Wagner model fails when cavitation effects appear and when a zone of pressure less than the ambient pressure reaches the contact line (Scolan & Korobkin 2015), which can occur when there is a strong deceleration during water entry.

3.1.2. Bagnold's solution. Bagnold's (1939) solution describes the impact of a liquid that traps a gas pocket and is based on gas pocket oscillations. The solution is characterized by the impact number S . Bagnold considered a 1D piston: A layer of incompressible liquid compresses a layer of perfect gas against a wall (**Figure 7**). The configuration shown in the figure has in addition a layer of gas on the other side of the liquid, which was not considered by Bagnold. We introduce the impact number $S = m_\ell V^2/(2 p_0 \Omega_0)$, where m_ℓ denotes the mass of the liquid, V the initial speed of the liquid, p_0 the initial pressure in the gas, and Ω_0 the initial volume of the entrapped gas. Note that the original impact number S introduced by Bagnold did not include the $1/2$ factor. The relation between the maximum pressure $p_{\max}$ at the wall and the impact number S is

$$\left(\frac{p_0}{p_{\max}}\right)^{1/\gamma} + \frac{1}{\gamma - 1} \left(\frac{p_{\max}}{p_0}\right)^{(\gamma-1)/\gamma} = S + \frac{\gamma}{\gamma - 1}, \quad 23.$$

where γ denotes the adiabatic exponent. The impact number S is the ratio between the initial kinetic energy of the liquid and the initial potential energy of the gas. For fixed acceleration, time

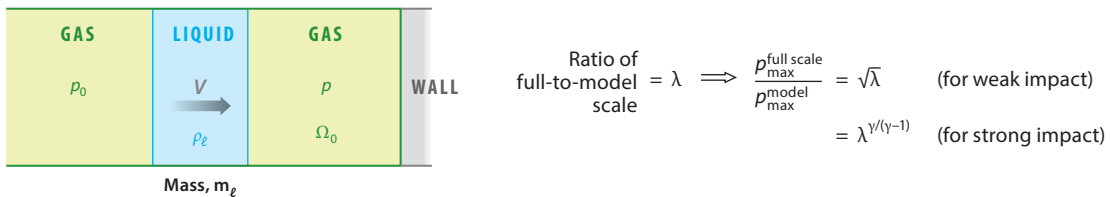


Figure 7

One-dimensional illustration of Bagnold's (1939) solution (elementary loading process ELP3): A liquid patch compresses a gas pocket, which then oscillates. For a full-to-model scale ratio λ , the full-to-model pressure ratio $p_{\max}^{\text{full scale}}/p_{\max}^{\text{model}}$ scales as $\sqrt{\lambda}$ for a weak impact and as $\lambda^{1/(\gamma-1)}$ for a strong impact. The additional layer of gas on the other side of the liquid was not considered by Bagnold.

Abbreviations: m_ℓ , the mass of the liquid; p_0 , the initial pressure in the gas; V , the initial speed of the liquid; Ω_0 , the initial volume of the entrapped gas.

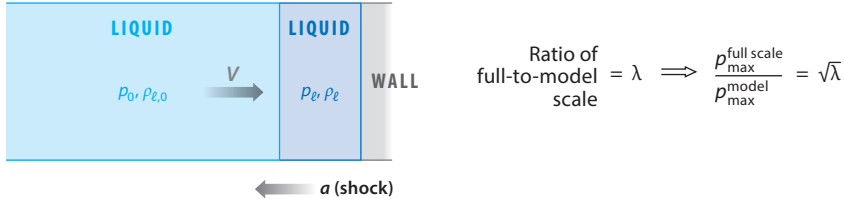


Figure 8

Illustration of the Rankine–Hugoniot solution. A shock wave propagates in the liquid after the compressible liquid impacts the right wall (elementary loading process ELP1). For a full-to-model scale ratio λ , the full-to-model pressure ratio $p_{\max}^{\text{full scale}}/p_{\max}^{\text{model}}$ scales as $\sqrt{\lambda}$.

scales as $\sqrt{\lambda}$. Hence, S scales as λ . Let θ be defined as $(p_{\max} - p_0)/p_0$. Then Equation 23 becomes

$$\left(\frac{1}{1+\theta}\right)^{1/\gamma} + \frac{1}{\gamma-1}(1+\theta)^{(\gamma-1)/\gamma} = \lambda S_1 + \frac{\gamma}{\gamma-1}, \quad 24.$$

where S_1 is the value of S for $\lambda = 1$. A study of Equation 24 gives the following relationship between θ and S : For small values of S , $(p_{\max} - p_0)/p_0$ scales as $\sqrt{\lambda}$, whereas for large values of S , $(p_{\max} - p_0)/p_0$ scales as $\lambda^{\gamma/(\gamma-1)}$ (e.g., $\lambda^{3.5}$ for air with $\gamma = 1.4$).

3.1.3. Rankine–Hugoniot solution. The shock waves involved in ELP1 can be understood through a 1D model that involves the compressibility of the liquid. It is based on the Rankine–Hugoniot conditions. We first use the isentropic model: Equation 4 supplemented with

$$\frac{\partial(\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) = 0, \quad 25.$$

and the EoS given in Equation 7. We address the situation in which the crest of the wave hits a rigid wall. If the liquid is initially (at the time of the impact) in a thermodynamic state (p_0, ρ_0) , and its normal velocity is $V = \mathbf{v}_0 \cdot \mathbf{n}$, then the Rankine–Hugoniot jump conditions stemming from Equations 7 and 25 are

$$\rho_{\ell,0} V = a(\rho_{\ell,0} - \rho_{\ell}), \quad \rho_{\ell,0} V^2 + p_0 - p_{\max} = a \rho_{\ell,0} V, \quad 26.$$

where $(p_{\max}, \rho_{\ell})$ defines the thermodynamic state of the liquid ($u_{\ell} = 0$), and a is the shock wave speed propagating in the liquid (**Figure 8**). Note that a and u_0 are of opposite sign because the liquid is reflected by the wall. Introducing the Mach number $\text{Ma}_0 = u_0/c$, where c is the constant speed of sound in Equation 7, $c^2 = p^{\text{ref}}/(\chi \rho^{\text{ref}})$, one can show that

$$p_{\max} = p_0 + \rho_{\ell,0} V c (1 + \mathcal{O}(\text{Ma}_0)), \quad a = -c(1 + \mathcal{O}(\text{Ma}_0)), \quad 27.$$

as Ma_0 tends toward zero. The relative pressure, $p_{\max} - p_0$, is close to the acoustic pressure $\rho_{\ell,0} V c$, and the shock speed is close to the speed of sound c in the liquid. The pressure scales as $\sqrt{\lambda}$. Hence, as far as scaling is concerned, we observe that the pressure impact due to a gas pocket (ELP3), modeled by Bagnold’s (1939) model, scales differently from that due to a direct liquid impact (ELP1) except for weak impacts (small S number). In Equation 27, we have invoked a weak-shock limit approximation.

3.1.4. Generalized Bagnold model. In the original Bagnold (1939) model, the liquid that compresses the gas is assumed to be incompressible. However, when the impact number S is large, the gas is so compressed that the liquid compressibility must then be taken into account. For large S , the maximum pressure is no longer given by Equation 23 but rather by the Rankine–Hugoniot

Impact number:
kinetic energy of the liquid divided by the potential energy of the gas;
 $S = m_{\ell} V^2 / (2 p_0 \Omega_0)$

REPRESENTATION OF THE INTERFACE

An attractive way to represent the interface is on meshes that move with the material of the two fluids. However, this is challenging for large deformations that distort the meshes. Eulerian fixed meshes are therefore mainly used when dealing with multifluid flows. We do not discuss the more sophisticated arbitrary Lagrangian–Eulerian methods (readers are referred to Donea et al. 2004 for more details). Hay et al. (2016) gave a recent example involving gas pocket and flip-through impacts. We also note that there is another direction that uses meshless methods such as SPH (see Section 3.2.5).

For multifluid flows, an appropriate method must be used to represent the interface. There are four main approaches. (a) In the first approach, used in potential flow solvers such as the boundary element method (see Section 3.2.4), the interface is the main unknown of the problem and must be found as part of the solution. (b) In the level set method, the following transport equation is solved for the interface:

$$\frac{\partial \varphi}{\partial t} + \mathbf{v} \cdot \nabla \varphi = 0,$$

where φ is a signed distance to the interface. This means that we must have $|\nabla \varphi| = 1$. Unfortunately, the discrete version of the above equation does not preserve this last condition, and some special techniques must be implemented (reviewed in Scardovelli & Zaleski 1999). Another drawback of this method is that mass is not conserved. (c) In the hybrid Lagrangian–Eulerian method, although the mesh is fixed, the interface is a line in two dimensions (or a plane in three dimensions) in each mixed cell that moves with the material (see Section 3.2.1). Mass conservation is ensured. (d) In the volume-of-fluid approach, the interface is captured rather than tracked. Here the volume fraction of each fluid α_k is computed in each cell with the transport Equation 28. This method does not lead to a sharp interface. Rather, the interface is located at the points where $\alpha_g = \alpha_\ell = 0.5$. This method is diffusive but certainly the easiest to handle from a numerical point of view. We note that antidiffusive correction terms can be used (such as in OpenFOAM); however, these are ad hoc techniques that depend on the model under consideration. Mass conservation is ensured. Section 3.2.3 discusses some codes using this approach.

solution (Equation 27). Thus, there are three regimes. For $S \leq 0.1$, the pressure scales as $\sqrt{\lambda}$; for $S \geq 4$, the pressure also scales as $\sqrt{\lambda}$; and for intermediate values of S , the pressure scales as $\lambda^{\gamma/(\gamma-1)}$. Full details are provided by Brosset et al. (2013).

3.2. Numerical Solutions

We now present numerical methods for the various models introduced in Section 2. We do not give an exhaustive list of the advantages and drawbacks of the various numerical methods, although we briefly comment on the suitability of the various methods. Instead, we emphasize how the local conservation laws are implemented in the solvers. For the main approaches to represent the interface, see the sidebar titled Representation of the Interface.

3.2.1. General. The general system of equations of Section 2.1 must be solved with a compressible multiphase flow solver such as the Flux-IC software (Braeunig et al. 2009b, Loubère et al. 2012). Flux-IC is a cell-centered, finite-volume platform designed for the computation of 2D and 3D compressible multimaterial flows. The interface is treated as a contact discontinuity between immiscible fluids (the hybrid Lagrangian–Eulerian method). In particular, phase transition is not taken into account. Both fluids are assumed to be compressible. Equations are discretized both

in time and in space in two or three dimensions. Time discretization is provided by the classical explicit Euler finite-difference scheme, whereas space discretization is performed on a structured mesh with rectangular control volumes (in two dimensions) by an upwind finite-volume method. Each control volume at a given time step can be either pure (i.e., filled with only one of the two fluids) or mixed (i.e., contains the two fluids but separated by an interface). In each cell, both fluids have their own characteristics (pressure, density, velocity, and EoS). The free-surface treatment allows the two fluids to be separated by a sharp interface—the two fluids can have relative tangential velocity when molecular viscosity is ignored.

3.2.2. Averaged models. As explained in Section 2.2, averaged models can hardly be avoided when the interface between the fluids becomes complicated. This is a very hard modeling problem. In fact, each regime might require a dedicated model. Let us give a few (simple) examples in the context of two immiscible fluids: (a) In homogeneous flows, both fluids are present everywhere with comparable volume presence. In general, Equations 14–17 are added to simplify the model with less equations. (b) In stratified flows, the gas is above the liquid. Equations 14–17 are satisfied, and the value of the exchange of momentum between the fluids ($\mathbf{M}_g$ and $\mathbf{M}_\ell$) will follow from this geometry. (c) In dispersed flows, one phase is said to be continuous and the other dispersed (e.g., bubbles in a continuous liquid or droplets in a continuous gas). In both cases, Equations 14–17 are satisfied, but depending on the situation, the values of $\mathbf{M}_g$ and $\mathbf{M}_\ell$ will again differentiate the regimes. For example, a droplet in a gas will experience a drag force depending on the relative velocity $V_g - V_\ell$. (d) For the disappearance of a phase, we assume that we start with a homogeneous flow and let gravity act. Then by decantation, the flow transitions toward a stratified flow. The exchange of momentum between the fluids ($\mathbf{M}_g$ and $\mathbf{M}_\ell$) takes into account this change of topology of the flow.

Computing transitions between regimes is still an outstanding question. Additionally, it turns out that the mathematical (and physical) properties of the model could be puzzling. For instance, nothing guarantees that exact Riemann solvers exist. (We refer readers to Toro 2010 concerning Riemann solvers in fluid mechanics.) This rules out all the numerical methods that rely on exact (or even approximate) Riemann solvers. Even worse, the convective part of average models might not be hyperbolic. Hyperbolicity is a necessary and sufficient condition for the evolutionary problem to be well posed (causality in physical words) in the absence of regularizing effects such as diffusion or surface tension (see Bresch et al. 2017). State-of-the-art numerical methods are based on finite-volume methods because exact conservation of mass, momentum, and energy is mandatory in the context of industrial simulations. Another important issue in this context is the presence of nonconservative products in the models. They come from the exchange term $\mathbf{M}_g$ in Equation 16. For instance, in the absence of diffusion, a minimal version of this equation is

$$\frac{\partial(\alpha_g \rho_g \mathbf{v}_g)}{\partial t} + \nabla \cdot (\alpha_g \rho_g \mathbf{v}_g \otimes \mathbf{v}_g) + \alpha_g \nabla p = \alpha_g \rho_g \mathbf{g}, \quad 28.$$

where the nonconservative product is $\alpha_g \nabla p$. At the discrete level, there are various ways to discretize such a term, and each one might lead to a different numerical solution. Ghidaglia et al. (2001) proposed a general solution. It consists of writing all the nonconservative products as follows:

$$\text{nonconservative product} = \sum_k \omega_k \frac{\partial f_j^k}{\partial x_j}, \quad 29.$$

where, for a given k , f_j^k is one of the components of the fluxes (mass or momentum), and ω_k is a calculable function of the dependent variables. Because in the finite-volume framework the dependent variables are known in the control volumes and the fluxes are known on the edges,

the discretization of $\omega_k(\partial f_j^k / \partial x_j)$ is no longer ambiguous. Bredmose et al. (2009, 2015) obtained numerical results based on averaged models. In particular, they confirmed that the strongest impact pressures occur when a small air pocket is trapped at the wall. They were able to model flip-through impacts and impacts trapping small air pockets. They found that the dividing line between flip-through with no trapped air and impacts that trap a small air pocket is not as sharp as potential flow computations suggest (see Section 3.2.4).

3.2.3. Incompressible Navier–Stokes equations. We briefly review the components of two widely used solvers: ANSYS Fluent and OpenFOAM. We simply describe the common procedure, knowing that there are other algorithms available. There are some other solvers such as STAR-CCM+ that we do not discuss here.

With regard to ANSYS Fluent, in the numerical model, the PISO (pressure-implicit with splitting of operators) algorithm is usually adopted to couple pressure and momentum equations, the body force weights scheme is used for pressure discretization, the geometric reconstruction scheme is used for the calculation of face fluxes for the volume-of-fluid (VOF) model, and the second-order upwind scheme is used for all other terms in the NS equations. In addition, the noniterative time advancement (NITA) scheme is employed to speed up transient simulations. The NITA scheme does not reduce the splitting error to zero for each time step but only makes it the same order as the truncation error. However, the NITA scheme still allows for an inner iteration to solve the individual set of equations. The details of these schemes can be found in the 15.0 release of the ANSYS Fluent User's Guide (ANSYS 2013). ANSYS Fluent has been used, for example, by Wei et al. (2016) to study the slamming of an OWSC.

With regard to OpenFOAM, the multiphase flow (gas and liquid) is solved with the VOF method. The numerical solution of the momentum equation is carried out by the PISO algorithm for the pressure–velocity coupling, and an implicit Euler method is used for time stepping. The VOF advection equation is solved by an explicit first-order time integration with a flux limiter on the divergence term. The OpenFOAM library uses a finite-volume approach for the discretization of the simulation domain. Gradient and divergence terms are computed using a second-order Gaussian integration, and a central differencing scheme is used for point-to-point interpolation. The details of these schemes can be found in the 3.0.1 OpenFOAM User Guide (OpenFOAM 2015). OpenFOAM has been used, for example, by Henry et al. (2014b) to study the slamming of an OWSC (the results have been compared with those obtained with the SPH method; see Section 3.2.5) and by Calderón-Sánchez et al. (2015) to study the free fall of a liquid block through gas.

Rafiee et al. (2013) used a finite-difference solver of the incompressible NS equations combined with the level set method for the tracking of an interface to conduct the International Society of Offshore and Polar Engineers (ISOPE) numerical comparative study described below (see also **Figure 5**). Because the results were compared with those obtained with the SPH method, they are discussed in Section 3.2.5.

3.2.4. Potential flow. The main advantage of potential flow theory is to solve for the velocity potential instead of the velocity $\mathbf{v}$ directly, assuming irrotational flow. From the physical point of view, small eddies are generated, for example, on the boundary of the solids. However, because the results obtained with potential flow solvers compare favorably with experimental results (Peregrine 2003), it is commonly assumed that rotational effects in the flow are negligible as far as slamming pressures are concerned. Brevig et al. (1982) developed a numerical method based on Cauchy's equation to study extreme wave forces on submerged WECs (the term slamming

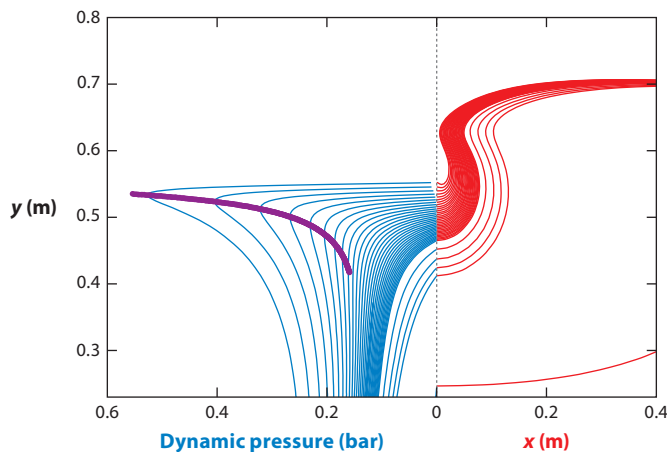


Figure 9

Potential flow simulations of a flip-through-type wave impact. (*Left*) Pressure distribution along the vertical wall. (*Right*) Free-surface profiles. The time increment between the profiles is equal to 0.035 s initially, 3.5 ms for the intermediate profiles, and 0.35 ms for the last profiles. Figure adapted with permission from Scolan (2010).

is mentioned twice). Interestingly, Cooker & Peregrine (1990) discovered the flip-through phenomenon numerically using a fully nonlinear potential flow solver. As shown in Section 4, the flip-through impact is easier to obtain numerically than experimentally.

A popular and efficient method for both 2D and 3D problems is the boundary element method, which is expected to work well when Froude scaling is appropriate (the water entry problem, for example). Readers are referred to Grilli et al.'s (2010) description of the boundary element method and Zhang et al.'s (1996) application to plunging waves impacting on a vertical wall.

A less popular method for 2D and 3D problems is the desingularized boundary integral method (Cao et al. 1991). Scolan (2010) combined this method with the use of conformal mapping, thus restricting it to 2D flows. Numerical results for a flip-through-type wave impact were provided, for example, by Bredmose et al. (2010) and Scolan (2010) (see **Figure 9**). Both papers mention the strong sensitivity of the impact type to the wave shape. 3D effects also lead to discrepancies in comparison with experimental results. The main weakness of the potential flow simulations is that the entrapment of air bubbles within the water body cannot be modeled.

3.2.5. Smoothed particle hydrodynamics. The SPH method is introduced above, but given that physics and numerics are intimately tied, we now discuss it in more detail. In recent years, the SPH method has emerged as an alternative approach to the more traditional, grid-based numerical methods when modeling violent impact problems (Monaghan 2012, Violeau 2012, Rafiee et al. 2012). The Lagrangian particle nature of SPH means that highly deforming flows undergoing severe geometry change are captured automatically, enabling all aspects of slamming to be described before, during, and after impact. Guilcher et al. (2013, 2014a,b) succeeded in developing an SPH solver with two compressible fluids (liquid and gas) interacting with the impacted structure through a complete coupling. The 2D numerical benchmark of a patch described below was tested. Lind et al. (2015) numerically modeled the rigid flat-plate impact using a two-phase SPH method, combining a compressible air phase with a truly incompressible water phase. Single-phase incompressible SPH can provide accurate pressure predictions for a wide range of free-surface incompressible flows, including water entry problems (Skillen et al. 2013) and sloshing (Gotoh

et al. 2014). Some of the two-fluid SPH methods are based only on weakly compressible SPH (see, e.g., Colagrossi & Landrini 2003, Gong et al. 2009, Colagrossi et al. 2010, Monaghan & Rafiee 2013) or purely incompressible formulations [see related work using the moving particle semi-implicit method (Khayyer & Gotoh 2013)]. Although water may remain incompressible, the severity of slamming requires a compressible air phase because air ejection velocities prior to impact can approach the speed of sound. Lind et al. (2015) imposed kinematics from the incompressible phase with pressure from the compressible phase at the interface, which enables a physical material discontinuity to be maintained at the interface, with the correct step change in density. The compressible phase is computed in a conventional explicit manner governed by an EoS.

3.2.6. Numerical benchmarks. Given the variety of numerical methods, numerical benchmarks are very useful to compare methods. At the ISOPE conference in 2010, GTT and University College Dublin conducted a numerical comparative study on the free fall of a liquid patch inside a tank filled with gas, with an update in 2013 (see **Figure 5**). The results were of mixed quality. With complete Froude scaling (i.e., including the scaling of the speeds of sound), there was excellent agreement with the analytical Bagnold (1939) solution for all participants, but there was a lot of disparity in the results for the 2D case, with a factor of 10 difference between the peak pressures. In 2016, ISOPE launched a numerical comparative study on the water entry problem (wedge drop and ship section drop) (Hong et al. 2017), and 18 out of 20 participants used an incompressible fluid model. Very good agreement was obtained for the peak pressure. However, the CFD results are quite sensitive to mesh size, and the authors concluded that an advanced grid convergence study was needed. Moreover, in the case of a tilted wedge, scattering was observed in the peak pressures. Elhimer et al. (2017) studied the impact between a rigid body and a gas–liquid mixture. They found a significant reduction of the impact load with the increase of the air volume fraction. In addition, several groups have been working on the breaking of waves on a vertical wall with most of the methods described above. **Figure 10** provides

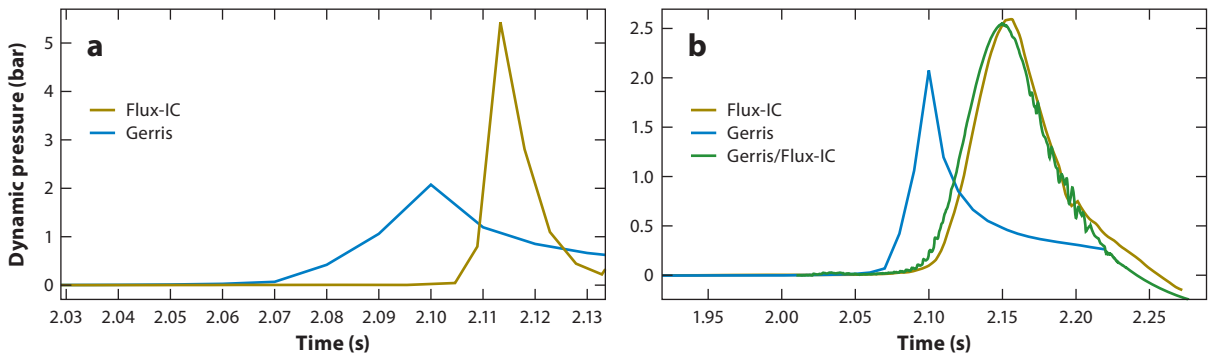


Figure 10

Wave pressure as a function of time for a wave breaking against a vertical wall. (*Left*) Comparison between incompressible (Gerris) and compressible (Flux-IC) approaches. (*Right*) Comparison between incompressible (Gerris), compressible (Flux-IC), and coupled (Gerris/Flux-IC) approaches. The pressure peak arrives earlier in the incompressible simulations and is smaller. Refining the resolution increases the pressure peak. The coupling between incompressible and compressible codes works well and leads to a significant gain in computation time. Readers are referred to Popinet (2003) for a description of Gerris. Figures adapted with permission from Mrabet (2017).

an illustration clearly showing the differences between compressible- and incompressible-flow approaches.

CFD codes are not suited for propagating waves but are suited for describing the impact pressures. Therefore, an interesting alternative is to couple different solvers, or at least to use one solver to generate an initial condition for another solver. For example, a potential flow solver can be used to generate overturning waves and then be used as an initial condition to a compressible multiphase flow solver (see **Figure 10**). Even with this type of approach, it is clear that CFD cannot be used to assess extreme impact pressures during slamming. The CPU time is orders of magnitude longer than real time (of the order of 10^4). Currently, the only way to assess such pressures is to perform model tests (Gervaise et al. 2009, Kuo et al. 2009). Note also that it is difficult to compare numerical and experimental impact pressures because of the variability of experimental pressures, even when the experimental setup allows for a perfect repetition of the global form of the wave before impact. This variability is due to free-surface instabilities, which are presently not well captured by CFD.

4. EXPERIMENTAL TECHNIQUES

Slamming experiments are challenging because of the short duration of slamming events and of their stochastic character. In the context of LNG sloshing, for example, model tanks instrumented by clusters of pressure sensors are placed on the platform of a hexapod and excited by six-degrees of freedom motions derived from full-scale seakeeping calculations scaled down using Froude scaling. Although, as stated above, benchmarks are regularly used to compare results obtained with various numerical flow solvers, the ISOPE sloshing committee introduced an experimental benchmark for sloshing in 2011 (Loysel et al. 2012). In the framework of a Joint Industry Project dedicated to the full-scale measurement of sloshing, one tank of an LNG carrier was equipped with several sensors, mainly strain gauges (Pasquier & Berthon 2012).

The duration of impacts is very short. At full scale, it is typically of the order of 10 ms or even less. Sampling rates of at least 20 kHz are recommended for fatigue damage studies (global) and up to 100 kHz for extreme loading studies (local) (Neugebauer et al. 2014). In addition to the requirements for the measurement system and the data-acquisition system, there is the problem of the pressure sensors themselves. Nowadays, there are piezoelectric and piezoresistive pressure sensors with very high natural frequencies. But the quoted high frequency is the frequency in air; the natural frequency for a sensor submerged in water is much lower. Kim et al. (2015) discuss some specific challenges. Scolan (2014a,b) recommended that, when comparing experimental pressures with numerical pressures, one averages the numerical pressures on the actual area of the pressure sensor.

4.1. Drop Tests

In a wet drop test, one equips a sample of the structure to be analyzed with pressure sensors and lets it fall freely onto calm water. The drop height is adjusted to study different impact velocities and drop incidences (the incidence is the relative angle between the surface of the structure and the liquid surface). Drop tests are repeatable for large deadrise angles of the wedge because the gas can escape almost freely without creating free-surface instabilities. In order to compare drop tests results with Wagner's (1932) solution (ELP2), the tests must be 2D. 3D tests always give smaller pressures because of edge effects (Zhao et al. 1997). As long as the free surface stays almost flat (before jet formation), the comparison between Wagner's solution and 2D tests is excellent. Drop tests are useful for assessing pressure sensors and providing values for the validation of CFD tools,

in particular incompressible ones. Air pockets at low angles of incidence tend to lower the impact pressure (Chung et al. 2007). When the angle of incidence increases, a jet is generated. Kang et al. (2008) investigated shallow-water impacts, and Oh et al. (2009) analyzed the generation process of air pockets during a flat impact of a box-type model. Elhimer et al. (2017) also found a reduction of the impact load when a cone enters a gas–liquid mixture due to the increase in the air volume fraction.

4.2. Sloshing Model Tests

Fluid flows generated in membrane tanks of LNG carriers or offshore LNG floating units may have different shapes, and sloshing impacts may occur at different locations, depending on the tank fill level, the magnitude of ship motion, and the range of dominant motion periods. For low fill levels, hydraulic bores and progressive waves move back and forth between the tank walls. The wave front may be inclined or almost vertical with an entrapped gas pocket. The impacted areas can cover all longitudinal walls. For a high fill level, standing waves move up and down and waves break in the tank corners. Sloshing experiments are usually conducted using a six-degree of freedom hexapod equipped with clusters of pressure sensors. In the past, no importance was given to the DR. Several studies led to the conclusion that keeping the same gas-to-liquid DR at both scales is a prerequisite for obtaining similar wave shapes before impact (Karimi et al. 2015). This is achieved at the model scale by using water and a mixture of N_2 and SF_6 . Because the speeds of sound cannot be Froude scaled (PFS), sloshing model tests introduce gas- and liquid-compressibility biases. Note that regardless of the sea conditions, the motion of the liquid in the LNG membrane tanks is always transverse to the ship's trajectory.

The first benchmark on sloshing model tests was carried out between 2011 and 2012 (Loysel et al. 2012). A second benchmark was conducted in 2013 (Loysel et al. 2013). These benchmarks aimed at comparing the testing equipment used by laboratories doing experiments on sloshing. To have a large number of participants and to make comparisons easier, the benchmark specified 14 simple conditions for a narrow rectangular tank, only for high fill levels (above 70% of the tank height).

Data from harmonic or irregular excitations were postprocessed to compare the two most important types of statistical results for a sloshing assessment: the number of impacts occurring during a prescribed time interval and the probability curves for pressure exceedance. Data from single wave impacts were simply used to compare the repeatability of the pressure results when impact conditions were perfectly repeated.

In addition to the sloshing model tests described above, several tests have been conducted by various groups across the world. For example, Lugni et al. (2006) focused on flip-through impacts, and Pistani & Thiagarajan (2012) analyzed sensors. Additionally, Lugni et al. (2014) studied the evolution of flip-through events upon a deformable wall during shallow-water sloshing, and Z. Wei et al. (2015) investigated sloshing-induced slamming in screen-equipped tanks in shallow-water conditions. Karimi et al. (2015, 2016, 2017) focused on the global and local effects of the gas-to-liquid DR. The DR clearly modifies wave shapes, as explained earlier.

4.3. Wave Impact Tests in Flume Tanks

Many experiments involving wave breaking against a vertical wall have been performed by the coastal engineering community (see the examples in Section 1). The generation of breaking waves in a coastal engineering flume by a wave-focusing method produces impacts that are similar to sloshing impacts in the membrane LNG tanks, the only difference being that rotations and induced

Coriolis accelerations lead to a greater variety of wave shapes in tanks. This observation led to the Sloshel experiments (see **Figures 1** and **2**). They consisted in large-scale experiments performed at Deltares (Delta flume and Schelde flume) by MARIN. At the same time, small-scale experiments were performed by GTT with Ecole Centrale de Marseille (MiniSlo 3 project) at two different scales. Impressive instrumentation was used in both cases (e.g., several hundred pressure sensors, high-speed cameras, strain gauges). Bogaert et al. (2010) and Kimmoun et al. (2010) compared impact pressures at two different scales from Sloshel and MiniSlo 3 experiments, respectively, and both concluded that Bagnold's (1939) gas pocket model worked quite well.

4.4. Slamming of Oscillating Wave Surge Converters

OWSCs are large buoyant flaps, hinged at the bottom of the ocean, that oscillate back and forth under the action of incident waves. They are intermediate-water-depth devices, and hence the interacting waves have an amplified horizontal motion, resulting in large amplitudes of flap rotation. In extreme sea states, as the flap pitches seaward and reaches a vertical position in the wave trough, the water line suddenly drops in front of the flap, and the flap hits the water surface with a high angular velocity, leading to a short-duration but high-magnitude load on the device (Henry et al. 2014b, Y. Wei et al. 2015). Wave slamming on an OWSC was first investigated in three dimensions by Henry et al. (2014b), using experimental and numerical methods. The slamming event was identified using images recorded by high-speed cameras (**Figure 11a**). It was found that it is the flap that impacts the water free surface rather than a wave hitting an object, like in classical

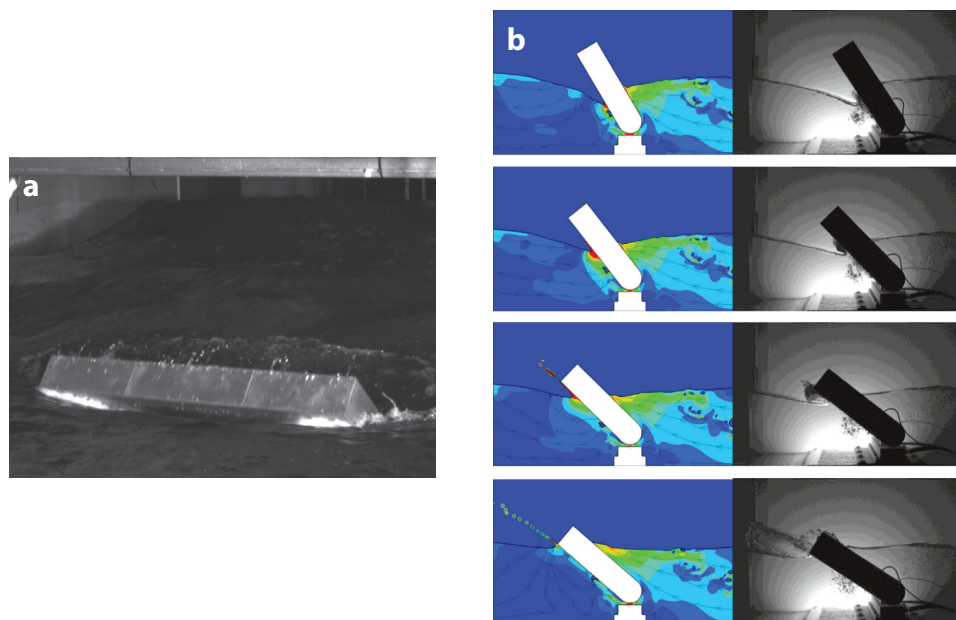


Figure 11

Slamming of oscillating wave surge converters (OWSCs). (a) Snapshot showing slamming in a three-dimensional experiment. Panel *a* reproduced with permission from Henry et al. (2014b). (b) Snapshot showing slamming in a two-dimensional experiment. The wave travels from left to right, and slamming occurs when the OWSC pitches seaward before the wave crest arrives. Panel *b* reproduced with permission from Wei et al. (2016).

wave impacts. Measurements by a finite number of pressure sensors indicated that the maximum pressure was located at the center of the flap. The slamming phenomenon and a sharp spike in the time history of the pressure were also captured numerically (Y. Wei et al. 2015). However, the spatial resolution in both experiments and numerical simulations was too low to capture the thin water jet and the local sharp pressure peaks accurately. Wave slamming phenomena have also been observed with the full-scale prototype of an OWSC (Oyster800) when it was operating in a rough sea state with a significant wave height greater than 5 m.

2D slamming experiments performed at Ecole Centrale de Marseille (Henry et al. 2014a, 2015) studied the progression of the jet up the face of the flap as it entered the water. Wei et al. (2016) provided comparisons with numerical results (**Figure 11b**). The main conclusion was that Wagner's (1932) model worked quite well, but compressible effects could also be clearly observed due to the presence of bubble swarms as the water rushed down and broke the free surface.

Slamming has also been observed in wave tank experiments on the spar-torus combination (STC) concept. The STC consists of a spar floating wind turbine and a torus-shaped heaving-body WEC. Due to the small draft of the torus (4 m), water exit and entry phenomena of the torus are expected. Significant slamming was observed experimentally when the WEC re-enters the water (Wan et al. 2015).

5. CONSEQUENCES ON THE DESIGN OF STRUCTURES

The final objective of slamming test campaigns is to check the ability of the structure (containment system, ship hull, floats, breakwaters, and WECs) to withstand the impact loads. This section discusses the consequences on the design of structures. This is a wide topic, and we concentrate here on the point of view of the classification societies because their primary objective is to verify the structural strength and integrity of essential parts of ship's hulls, LNG tanks, and WECs.

5.1. Ship Slamming

Guides from classification societies contain a description of the direct slamming strength assessment procedure for bottom slamming, bow-flare slamming, and stern slamming. The procedure consists of slamming load prediction and strength assessment (see, e.g., Am. Bur. Shipp. 2011). The slamming pressure can be obtained from either experiments or numerical simulations. The popular approach is the 2D slamming approach, which is focused on 2D slamming analysis using short-term extreme velocities and local pressure coefficients. An advanced approach may also be applied to obtain the slamming pressure, such as 3D slamming model tests or CFD simulations, which is called a 3D slamming approach. The expressions for the maximum pressure are similar to Equation 22. Note that the classification society DNV GL uses a 3D approach (Det Norske Veritas 2014).

5.2. Liquefied Natural Gas

Violent sloshing motions may occur, directly resulting in the occurrence of different impact conditions, which can lead to large structural loads possibly damaging both the cargo containment system and the inner hull structure (Am. Bur. Shipp. 2009, Lloyd's Regist. 2009, Bur. Veritas 2011a). Sloshing phenomena are complex from a thermodynamic point of view. Partial condensation of the LNG vapor (depending on the speed of condensation and impact duration) may happen during the impacts (Yung et al. 2010, Ancellin et al. 2012). Given the difficulties related to the correct evaluation of fluid-structure interactions during impacts in the tanks of LNG

carriers or offshore LNG floating units, the direct (absolute) approach is not yet feasible (Gervaise et al. 2009), and the so-called comparative approach is used. Loads or structural responses of new designs are compared with the reference case, which represents the LNG carrier designs and operations proven acceptable through service. The description of the reference case is given in guides provided by classification societies.

The point of view of most classification societies is that the evaluation of LNG impact pressures by CFD analysis is not yet reliable. High-impact pressure is strictly localized in space and time, being very sensitive to the local effects. Thus, it depends on many physical parameters of the liquid, gas, and structure involved in the impact. Hence, classification societies suggest that researchers use CFD analyses to identify the type and nature of the impact to accurately evaluate the kinetic energy of the liquid and quantify the impact by the impact normal velocity with respect to the wall at distinct locations (predefined hot spots).

Structural impact tests and dynamic finite-element analyses show that impact peak pressures are filtered and averaged by the containment system when the impulse reaches the steel back-up structure. Thus, the quasi-static pressure directly calculated by CFD is provided for the structural assessment of the inner-hull structure, as described by Bur. Veritas (2011b). Nevertheless, the peak pressure is taken into account by amplification of the quasi-static pressure by a partial safety factor.

One validation of CFD tools is against drop tests (Tenzer et al. 2015). For wedges with deadrise angles greater than 5°, Wagner's (1932) solution for pressure has proven to be satisfactory. Thus, several simulations at different heights are carried out, and the measured time pressure histories are compared to Wagner's reference solution. If the difference is too high, the CFD tool should be rejected. This validation looks primarily at incompressible effects. Neugebauer et al. (2014) validated NS equations solvers (incompressible liquid, compressible gas) by comparing numerical results with experimental results of single wave impacts inside a 2D tank.

5.3. Wave Energy Converters and Offshore Wind Turbines

There is a parallel between the strong impacts that can occur inside the tanks of LNG carriers and those that can occur on WECs (Tiron et al. 2015). Two aspects must be taken into consideration. First, for flap-type WECs, the flap can slam onto the water surface in extreme sea conditions. Second, for all types of WECs for which waves are likely to break on the site of the device, wave loads from breaking waves must be considered in the design of the structure. Wave loads from breaking waves depend on the type of breaking waves due to differences in the wave kinematics. For plunging waves, an impact load model may be used to calculate the forces on the structure. The impact pressure from a plunging wave can be expressed as $p_{\max} = \frac{1}{2} C_s \rho_\ell V^2$, where V denotes the water particle velocity at the plunging wave crest and C_s is the slamming coefficient. For a smooth cylinder, the slamming coefficient should not be less than 3.0. The upper limit for the slamming coefficient is 2π . For spilling breakers, the total load can be calculated using a strip theory in which the slamming coefficient for each strip is a function of the submergence of the strip, decaying from 5.15 when the wave hits the strip to 0.8 when the strip is fully submerged.

As indicated by Coe & Neary (2014), a group of researchers at Ghent University conducted slamming analyses of a composite WEC point absorber (Van Paepegem et al. 2011). Large-scale physical drop tests were performed in conjunction with finite-element analysis modeling to determine loading within the structure. CFD simulations of the slamming event were also conducted. As the use of composite structures in this application is somewhat novel, tests were specifically designed to assess the effects of reduced structural rigidity. For the device tested, the increased ductility of the composite structure reduced pressure loads by roughly 50%. The results

Table 1 Scaling for impact pressures based on the ELP

	ELP1 (direct impact)	ELP2 (building jet)	ELP3 (gas pocket)	
Analytical model	Rankine–Hugoniot compressible liquid ^a	Wagner incompressible liquid ^b	Bagnold compressible gas ^c	Generalized Bagnold compressible liquid and gas
Scaling (PFS)				
Weak impact	$\sqrt{\lambda}$	λ	$\sqrt{\lambda}$	$\sqrt{\lambda}$
Moderate impact	$\sqrt{\lambda}$	λ	NA	$\lambda^{\gamma/(\gamma-1)}$
Strong impact	$\sqrt{\lambda}$	λ	$\lambda^{\gamma/(\gamma-1)}$	$\sqrt{\lambda}$

The scaling shown in the table assumes that the same fluids are used at full and model scale. Abbreviations: ELP, elementary loading process; γ , the adiabatic constant; λ , the ratio of the full to model scale; NA, not applicable; PFS, partial Froude scaling.

^aRankine (1870), Hugoniot (1887, 1889).

^bWagner (1932).

^cBagnold (1939).

based on high-fidelity models showed major discrepancies with values determined using semi-empirical methods suggested by DNV GL. Clearly, the norms for WECs are still at their infancy and will have to be improved in the future.

Similarly, slamming phenomena are not yet sufficiently taken into consideration in the design of both fixed and floating offshore wind turbines. Slamming not only influences vertical loads but, more importantly, also changes the horizontal wave loads on the piles of fixed turbines.

6. SUMMARY AND DISCUSSION

In experimental methods and engineering applications, scaling is a major difficulty because all the physical phenomena cannot be scaled simultaneously in general. The usual approach consists in (a) selecting what is considered to be the main phenomena, (b) writing the corresponding model, and (c) ensuring that the equations in dimensionless form are the same at both scales. There are now two questions. First, is it possible to find fluids that fulfill step c? In general the answer is “no,” which leads to investigations of the bias induced by this absence of scaling. Second, as far as the full-scale application is concerned, was step a done correctly? In other words, have we selected all the necessary phenomena?

For PFS, the scaling of the impact pressure due to slamming depends on the ELP. A summary is given in **Table 1**. Numerical methods (CFD) can capture impulses but cannot properly capture slamming pressure peaks because they are rare events. This aspect is common to all problems related to rare events. Indeed, as observed in experiments, statistical convergence is very hard to achieve for such events (Karimi et al. 2017). Several hours or even days are required. In CFD, meaningfully computing one minute of physical time generally requires hours even on high-power computers. Another issue in our context is that the complete modeling of the phenomena must rely on complicated models (turbulence, capillarity, rotational flows, phase change), making even a single accurate computation a true challenge at the level of the present state of the art in CFD. Moreover, surface instabilities add much variability. But CFD is a valuable tool for at least the following reasons: All values of the physical parameters can be tested, physical hypotheses or simple models can be validated, physical phenomena can be segregated, and experiments that would be almost impossible to perform for practical reasons can be conducted via CFD. It follows that only if we work consistently in the three directions—theoretical, experimental, and numerical—can we

answer the question of estimating loads caused by slamming. All three models have to be nested to approach, if not access, the full physical details.

SUMMARY POINTS

1. Several mathematical models of different levels of complexity are used to evaluate impact pressures during slamming. Incompressible flow models can be used for loading processes that follow the Froude law of similarity. When bubbles or droplets form, averaged models can be used. Compressible multiphase flow models must be used for all the other cases.
2. Asymptotic solutions can be obtained for ELPs that can occur during slamming. The variability complicates the experimental discrimination of the effect on impact loads of the parameters (e.g., DR, gas compressibility, liquid compressibility, phase transition).
3. Numerical solutions can be obtained for different models with different approaches for tracking the interface. Even with coupling different solvers based on different models, CFD cannot yet assess impact pressures during slamming because the CPU time is at least four orders of magnitude longer than real time. However, CFD analyses are useful to identify the type and nature of impacts.
4. Experimental techniques are valuable in the study of slamming. The variability of local pressure measurements during liquid impacts is paramount. The local variability of the loads results mostly from the free-surface roughness and the liquid fragmentation resulting from the free-surface instabilities that develop during the escape of the gas.
5. In terms of the design of structures, the direct (absolute) approach is not yet feasible: In particular, it would require one to perform model tests with scaled physical properties like viscosity, speed of sound, and surface tension. Instead, a comparative approach is used based on pressure measurements in both existing ships and mock-ups of ships.

FUTURE ISSUES

1. Because tanks of LNG carriers now have a variety of sizes and are used with a variety of fill levels, elastic effects in the structure could lead to resonance phenomena.
2. Phase transition between the liquid and vapor is an important phenomenon that occurs during impacts in LNG tanks. It is presently not modeled at the small scale.
3. Scaling remains the most important open question for all slamming applications. Impact pressures are strongly influenced by local phenomena such as local geometry (raised edges of the cargo containment system), gas/liquid mixture (DR between the gas and the liquid, gas and liquid compressibilities), and surface tension. Surface tension effects, which are quite subtle, have been largely ignored until now. Indeed, surface tension affects slamming pressures not directly but indirectly. Surface tension leads to drastic changes in the geometry of the free surface (Frihat et al. 2017).
4. Slamming belongs to the class of extreme phenomena, such as freak waves. Extreme phenomena usually follow fat-tailed probability distributions. A full review article could be written on the topic of statistics/probability related to slamming.

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